

ALOFI, Niue

WMO: 918220

Lat: 19.067S

Lon: 169.917W

Elev: 72

StdP: 14.66

Time Zone: -11.00 (W11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	60.6	62.9	55.4	65.6	67.7	57.2	70.0	68.8	24.6	73.5	22.4	73.8	1.6	150	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.1	87.4	78.8	86.3	78.4	85.9	78.2	80.7	84.4	80.0	83.8	79.5	83.4	9.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	152.0	82.7	79.0	150.8	82.5	78.6	148.9	82.2	44.3	85.3	43.7	84.3	43.1	83.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.4	19.3	17.5	DB	57.9	88.9	3.1	0.9	55.6	89.5	53.8	90.1	52.1	90.6	49.8	91.2	(4)
(5)			WB	55.4	82.5	2.4	1.2	53.7	83.3	52.4	84.0	51.0	84.7	49.3	85.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.6	79.3	80.1	79.7	78.5	76.1	74.3	73.4	73.0	74.2	75.4	76.9	78.6
	DBStd	3.39	1.92	1.67	1.89	1.79	2.11	2.72	3.22	2.80	2.48	2.30	2.47	2.08
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9714	909	844	920	854	810	730	725	714	727	787	808	886
	CDD65	4239	444	424	455	404	345	280	260	249	277	322	358	421
	CDH74	29291	3850	3919	4062	3120	2029	1312	1103	1010	1220	1776	2463	3427
CDH80	5680	907	1005	1116	666	228	120	73	83	102	234	437	709	
Wind	WSAvg	7.6	7.6	7.7	6.0	6.8	7.7	7.5	6.8	8.2	8.9	8.5	8.2	7.8
Precipitation	PrecAvg	82.0	11.3	8.6	10.8	9.4	6.6	4.3	3.7	4.1	4.7	5.3	6.1	7.1
	PrecMax	125.4	38.3	20.1	35.4	19.5	17.1	21.1	10.7	14.4	14.2	13.5	14.4	22.9
	PrecMin	33.0	2.4	0.6	1.0	0.3	0.3	0.6	0.3	0.2	0.6	0.3	0.2	0.9
	PrecStd	21.2	8.0	5.0	7.2	5.0	4.0	4.3	2.8	4.0	3.7	3.6	3.6	5.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.0	88.1	88.2	86.9	84.6	84.3	83.7	83.9	84.1	85.1	86.3	87.5
		MCWB	79.2	78.9	79.3	78.2	78.1	77.2	76.1	75.7	76.1	76.4	77.5	77.9
	2%	DB	86.4	87.1	87.0	86.0	83.6	82.7	82.1	82.4	82.5	84.0	84.6	86.1
		MCWB	78.6	78.7	78.9	78.0	77.2	76.2	75.4	74.8	75.3	76.1	77.0	77.7
	5%	DB	85.8	86.1	86.2	84.6	82.4	81.3	80.6	80.8	80.9	82.5	83.9	84.6
		MCWB	78.3	78.5	78.6	77.3	76.2	75.2	74.4	73.8	74.2	75.1	76.7	77.1
	10%	DB	84.4	85.2	85.4	84.0	81.0	80.3	79.1	79.1	79.8	81.0	82.6	83.9
		MCWB	77.3	78.1	78.2	77.1	74.7	74.4	73.1	72.8	73.5	74.1	75.9	76.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	81.4	81.4	80.9	80.0	78.7	78.6	78.0	78.2	78.9	79.9	80.1
		MCDB	84.7	84.7	85.1	84.0	82.8	82.2	81.0	81.2	81.7	82.4	83.2	83.7
	2%	WB	80.1	80.2	80.5	79.8	78.8	77.4	77.1	76.4	76.9	77.8	79.0	79.4
		MCDB	84.0	84.2	84.6	83.3	81.6	81.0	80.2	80.1	80.3	81.8	82.7	83.4
	5%	WB	79.4	79.7	79.8	79.0	77.8	76.5	76.0	75.4	75.9	76.8	78.2	78.7
		MCDB	83.3	83.7	84.1	82.7	80.8	80.0	79.1	79.3	79.5	81.0	82.0	83.0
	10%	WB	78.7	79.1	79.1	78.3	76.7	75.7	74.9	74.3	74.8	75.8	77.2	78.0
		MCDB	82.7	83.2	83.5	82.0	79.8	79.0	77.9	78.1	78.3	79.5	81.3	82.1
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.1	11.4	10.7	10.7	10.8	11.2	11.4	10.7	11.1	11.0	10.3
		MCDBR	11.5	11.5	12.4	12.3	12.0	12.4	12.3	13.7	12.4	13.2	12.2	11.6
		MCWBR	5.7	5.5	6.4	7.0	7.2	7.4	7.3	8.0	7.1	7.0	6.4	5.9
	5% WB	MCDDBR	10.0	10.0	11.0	10.9	11.0	10.6	10.4	11.7	11.0	11.5	11.5	10.3
		MCWBR	5.7	5.9	6.5	7.5	7.8	7.2	7.0	7.9	7.4	7.2	7.0	6.2
Clear-Sky Solar Irradiance	taub	0.386	0.383	0.377	0.386	0.380	0.378	0.368	0.374	0.395	0.383	0.380	0.384	
	taud	2.623	2.631	2.663	2.622	2.617	2.601	2.624	2.602	2.543	2.581	2.613	2.607	
	Ebn at Noon	304	301	296	281	271	266	272	281	286	298	304	305	
	Edh at Noon	32	32	30	29	27	27	27	29	33	33	33	33	
All-Sky Solar Radiation	RadAvg	1959	1922	1787	1543	1296	1201	1280	1486	1690	1887	1953	2030	
	RadStd	200	189	141	113	102	81	61	90	101	135	134	140	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page